

# Complete OR Formulation — Hauler Route Optimization

Scope: Multi-stop Factory → Delivery Centre → Factory routing with hauler continuity, driver assignment, 11-hour driver limits, multi-driver trips, handover, capacity, timing and operational constraints.

Primary variables:  $y_{ijk}$  selects the hauler movement;  $x_{ijkl}$  assigns the driver to that selected movement. Auxiliary variables are used only where required.

## 1. Problem definition

A complete trip starts at a factory, visits one or more delivery centres, and returns to a factory. A trip may be less than or greater than 11 hours. The 11-hour limit applies to each driver, so a longer trip can remain feasible by using multiple drivers.

Complete trip = Factory → DC<sub>1</sub> → DC<sub>2</sub> → ... → DC<sub>n</sub> → Factory

Example: F1 → D1 → D2 → F1, with Driver 1 operating F1 → D1 → D2 and Driver 2 taking over at D2 for D2 → F1.

## 2. Sets

H = set of haulers / trucks.

F = set of factories.

D = set of delivery centres.

N = F ∪ D = all locations.

R = set of drivers / pilots.

## 3. Parameters

Q<sub>i</sub> = capacity of hauler i.

q<sub>d</sub> = demand at DC d.

d<sub>jk</sub> = distance j → k.

τ<sub>jk</sub> = travel time j → k.

c<sub>ijk</sub> = hauler transportation cost for i on j → k.

C<sub>ijkl</sub><sup>D</sup> = driver cost for l operating i on j → k.

s<sub>j</sub> = service time at j.

A<sub>jk</sub> ∈ {0,1} = allowed movement indicator.

Compat<sub>il</sub> ∈ {0,1} = driver-hauler compatibility.

E<sub>d</sub>, L<sub>d</sub> = earliest/latest arrival at d.

T<sub>i</sub><sup>max</sup> = maximum complete-trip duration, if applicable.

D<sub>i</sub><sup>max</sup> = maximum trip distance, if applicable.

R<sub>i</sub><sup>max</sup> = maximum drivers on trip i, if applicable.

h<sub>k</sub> ∈ {0,1} = whether handover is allowed at k.

H<sub>max</sub> = driver duty/driving limit = 11 hours.

## 4. Decision variables

Hauler route:  $y_{ijk} \in \{0,1\}$ . It equals 1 if hauler i travels from j to k. Example:  $y_{H1,F1,D2} = 1$ .

Driver assignment:  $x_{ijkl} \in \{0,1\}$ . It equals 1 if driver l operates hauler i from onboarding j to offboarding k. Example:  $x_{H1,D2,D4,D7} = 1$ .

Auxiliary: a<sub>if</sub> for origin factory; v<sub>id</sub> for DC visit; z<sub>il</sub> for driver used on a trip; T<sub>ij</sub> for arrival time; L<sub>ij</sub> for load; u<sub>id</sub> for route order; optional interval variables for scheduling.

## 5. Objective function

$$\min Z = \sum_{i,j,k} C_{ijk} y_{ijk} + \sum_{i,j,k,l} C_{ijkl}^D x_{ijkl} + \text{Penalties}$$

The first term represents physical hauler/route cost; the second represents driver cost. Penalties may cover late/unserved demand, waiting, overtime or extra resources. Example: 100 km  $\times$  ₹20/km = ₹2,000 through y; 5 h  $\times$  ₹500/h = ₹2,500 through x.

## 6. Core routing constraints

### 6.1 Allowed route

$$y_{ijk} \leq A_{jk}$$

Prohibited arcs cannot be selected. Example:  $A_{F1,D3}=0 \Rightarrow y_{H1,F1,D3}=0$ .

### 6.2 One factory departure

$$\sum_{f \in F} \sum_{k \in N} y_{ifk} = 1 \quad \forall i$$

Each hauler starts one complete trip.

### 6.3 One factory return

$$\sum_{j \in N} \sum_{f \in F} y_{ijf} = 1 \quad \forall i$$

Each hauler finishes by entering a factory.

### 6.4 Same factory return

$$\sum_f a_{if} = 1; \sum_k y_{ifk} = a_{if}; \sum_j y_{ijf} = a_{if}$$

Forces origin and return to be the same factory. Remove/modify if cross-factory return is allowed.

### 6.5 Truck continuity

$$\sum_{j \in N} y_{ijk} = \sum_{m \in N} y_{ikm} \quad \forall i, k \in D$$

Every hauler arrival at an intermediate DC is followed by one departure. Example: F1  $\rightarrow$  D1 selected  $\Rightarrow$  one outgoing movement from D1.

### 6.6 DC coverage

$$\sum_{i \in H} \sum_{j \in N} y_{ijd} = 1 \quad \forall d \in D$$

Each mandatory DC is served exactly once when split delivery is not allowed.

### 6.7 Visit indicator

$$v_{id} = \sum_{j \in N} y_{ijd}$$

Links the route to whether hauler i serves DC d.

### 6.8 No repeated DC

$$\sum_{j \in N} y_{ijd} \leq 1 \quad \forall i, d$$

Prevents repeated visits by the same hauler.

### 6.9 No self-loop

$$y_{ijj} = 0$$

Prohibits  $j \rightarrow j$ .

### 6.10 Subtour elimination

$$u_{id} - u_{ik} + |D| y_{idk} \leq |D| - 1$$

Prevents disconnected DC-only cycles. In CP-SAT, a circuit/path constraint is often cleaner.

## 7. Route-driver coupling

### 7.1 Exactly one driver on each selected segment

$$\sum_{l \in R} x_{ijkl} = y_{ijk} \quad \forall i, j, k$$

If  $y=0$ , no driver is assigned; if  $y=1$ , exactly one driver operates the segment.

### 7.2 Driver-hauler compatibility

$$x_{ijkl} \leq \text{Compat}_{il}$$

An unqualified driver cannot operate that hauler.

### 7.3 Driver-trip indicator

$$x_{ijkl} \leq z_{il}; \sum_l z_{il} \leq R_i^{\max}$$

Identifies drivers used on the complete trip and controls crew size.

## 8. Capacity and demand

### 8.1 Hauler capacity

$$L_{ij} \leq Q_i$$

Load cannot exceed capacity. Example: 20T truck cannot carry 25T.

### 8.2 Split delivery allowed

$$\sum_{i \in H} q_{id} = \text{Demand}_d$$

Multiple haulers can jointly satisfy a DC's demand.

### 8.3 Split delivery not allowed

$$\sum_{i \in H} v_{id} = 1; q_{id} = \text{Demand}_d v_{id}$$

One hauler supplies the complete demand.

### 8.4 Load conservation

$$L_{id}^{\text{out}} = L_{id}^{\text{in}} - q_{id}$$

The load decreases by the quantity delivered at each DC.

## 9. Time and delivery

### 9.1 Travel-time propagation

$$T_{ik} \geq T_{ij} + s_j + \tau_{jk} - M(1 - y_{ijk})$$

Selected movement respects service plus travel time.

### 9.2 Delivery window

$$T_{id} \geq E_d - M(1 - v_{id}); T_{id} \leq L_d + M(1 - v_{id})$$

Example: 10:00-14:00 window  $\Rightarrow$  arrival must be inside that interval.

### 9.3 Service time

$$\text{Departure}_{id} = \text{Arrival}_{id} + s_d$$

Unloading/loading time is included in the schedule.

### 9.4 Factory hours

$$E_f \leq \text{StartTrip}_i \leq L_f$$

Departure/loading must respect factory operating hours.

## 10. 11-hour and multi-driver logic

### 10.1 Individual driver limit

$$\sum_{i,j,k} T_{ijk}^{\text{duty}} x_{ijkl} \leq 11 \quad \forall l$$

Each driver's assigned duty/driving time is at most 11 hours. Confirm whether waiting/service counts.

### 10.2 Long complete trips

$$\text{TripDuration}_i \leq 11 \cdot \sum_{l \in R} z_{il}$$

Example: 8 h  $\Rightarrow$  at least 1 driver; 17 h  $\Rightarrow$  at least 2; 23 h  $\Rightarrow$  at least 3.

### 10.3 Interpretation

$\text{TripDuration}_i > 11$  does not imply infeasibility

The trip remains feasible if its portions can be covered by available drivers respecting the 11-hour limit and handover rules.

## 11. Driver handover

### 11.1 Physical handover

$$x_{i,j,k,l}=1 \text{ and } x_{i,k,m,r}=1 \Rightarrow \text{driver } l \text{ hands over hauler } i \text{ to driver } r \text{ at } k$$

The common point  $k$  maintains physical continuity.

### 11.2 Allowed handover points

$$\text{Handover}_{ik} \leq h_k$$

No handover at locations where  $h_k=0$ .

### 11.3 Handover duration

$$\text{Start}_{\text{next}} \geq \text{End}_{\text{previous}} + \text{HandoverTime}_k$$

Allows time to complete the driver transfer.

### 11.4 Driver availability

$$\text{Start}_{ijkl} \geq A_l - M(1 - x_{ijkl}); \text{End}_{ijkl} \leq B_l + M(1 - x_{ijkl})$$

Assignment must fit the driver's availability.

### 11.5 Driver non-overlap

$$\text{End}_a \leq \text{Start}_b \text{ OR } \text{End}_b \leq \text{Start}_a$$

A driver cannot operate two haulers simultaneously; CP-SAT interval + NoOverlap is recommended.

### 11.6 Driver repositioning

$$\text{Start}_{\text{next}} \geq \text{End}_{\text{previous}} + \tau_{p,q}^{\text{driver}}$$

Driver must have time to move from  $p$  to  $q$  for the next assignment.

### 11.7 Driver skill

$$x_{ijkl} \leq \text{SkillCompat}_{l,j,k}$$

Required route/vehicle/product/customer certification must be satisfied.

### 11.8 Break/rest

$$\text{ContinuousDrivingTime}_l \leq B_l; \text{StartNext} \geq \text{EndPrevious} + \text{RestTime}$$

Add if required by regulation/customer policy.

## 12. Hauler operational constraints

### 12.1 Vehicle/DC compatibility

$$v_{id} \leq \text{Compat}_{id}$$

Only eligible haulers serve a DC.

### 12.2 Maximum DCs

$$\sum_{d \in D} v_{id} \leq K$$

Example:  $K=3$  allows at most three delivery centres.

### 12.3 Maximum complete-trip duration

$$\text{TripDuration}_i \leq T_i^{\text{max}}$$

Separate from the 11-hour driver rule. Example: 18 h trip is allowed if max trip duration is 24 h.

### 12.4 Maximum distance

$$\sum_{j,k} d_{jk} y_{ijk} \leq D_i^{\text{max}}$$

Optional vehicle/trip distance limit.

## 12.5 Hauler availability

$$\text{StartTrip}_i \geq A_i; \text{EndTrip}_i \leq B_i$$

Trip must fit the hauler availability.

## 12.6 Road/vehicle restrictions

$$y_{ijk} \leq \text{Allowed}_{ijk}$$

Captures weight, height, hazardous-material and vehicle-type restrictions.

## 13. Business/service constraints

### 13.1 Mandatory delivery

$$\sum_{i,j} y_{ijd} = 1$$

Every mandatory DC is served.

### 13.2 Optional delivery

$$\sum_{i,j} y_{ijd} + u_d = 1$$

$u_d = 1$  means unserved; penalize it in the objective.

### 13.3 Priority

Objective includes  $\text{Penalty}_d \cdot u_d$

Critical DCs receive larger penalties.

### 13.4 Deadheading

Load may be 0 on  $y_{ijk} = 1$  if empty movement is allowed.

Confirm whether empty travel is permitted and how it is costed.

### 13.5 Dedicated hauler

$$v_{id} = 1 \Rightarrow v_{id'} = 0 \text{ for } d' \neq d$$

Optional if a customer requires a dedicated hauler.

## 14. How y and x work together

$y_{ijk} \rightarrow$  WHERE the hauler moves

$x_{ijkl} \rightarrow$  WHO operates the hauler on that movement

$\sum_l x_{ijkl} = y_{ijk} \rightarrow$  links route and driver decisions

Example: route  $F1 \rightarrow D1 \rightarrow D3 \rightarrow F1$  is selected by  $y$ .  $x$  can assign Driver 1 to the first two segments and Driver 2 to  $D3 \rightarrow F1$  when the 11-hour rule requires a handover.

## 15. Why no trip index t is needed

No  $t$  is required if each hauler performs exactly one complete Factory  $\rightarrow$  DC(s)  $\rightarrow$  Factory trip in the planning horizon.

$$\sum_{f,k} y_{ifk} = 1$$

$$\sum_{j,f} y_{ijf} = 1$$

These constraints define one closed trip per hauler. If the same hauler can perform multiple independent complete trips in the same planning horizon, introduce  $t$  and use  $y_{ijkt}$  and  $x_{ijklt}$ .

## 16. Critical customer clarifications

- Does 11 hours mean driving time or total duty time?
- Are drivers sequentially taking over the hauler, or can multiple drivers be onboard simultaneously?
- Where can handover occur, and how long does it take?
- Must the hauler return to the same factory?
- Can one hauler visit multiple DCs, and what is the maximum number?

- Is split delivery allowed?
- Can drivers reposition independently between assignments?
- Can the same hauler make multiple complete trips in the planning horizon?
- Is there a maximum complete-trip duration despite allowing >11-hour trips?
- Are empty/deadhead movements allowed?
- What should be minimized: cost, distance, time, number of haulers, number of drivers, service failures, or a weighted combination?

## 17. Final compact formulation

Objective

$$\min Z = \sum c_{ijk} y_{ijk} + \sum C_{ijkl}^D x_{ijkl} + \text{penalties}$$

Route

$y \leq A$ ; one factory departure; one factory return; flow conservation; DC coverage; no repeat; subtour elimination.

Driver

$\sum x = y$ ;  $x \leq \text{Compat}$ ; individual driver duty  $\leq 11$ ; availability; non-overlap; repositioning; handover.

Operations

$\text{Load} \leq \text{capacity}$ ; demand fulfilment; time windows; travel-time propagation; vehicle/DC compatibility; maximum stops/distance/trip duration where required.

Recommended CP-SAT architecture:  $y$  for route/arc selection,  $x$  for driver assignment, and auxiliary time/load/Boolean/interval variables for feasibility. Use circuit/path and NoOverlap where appropriate.